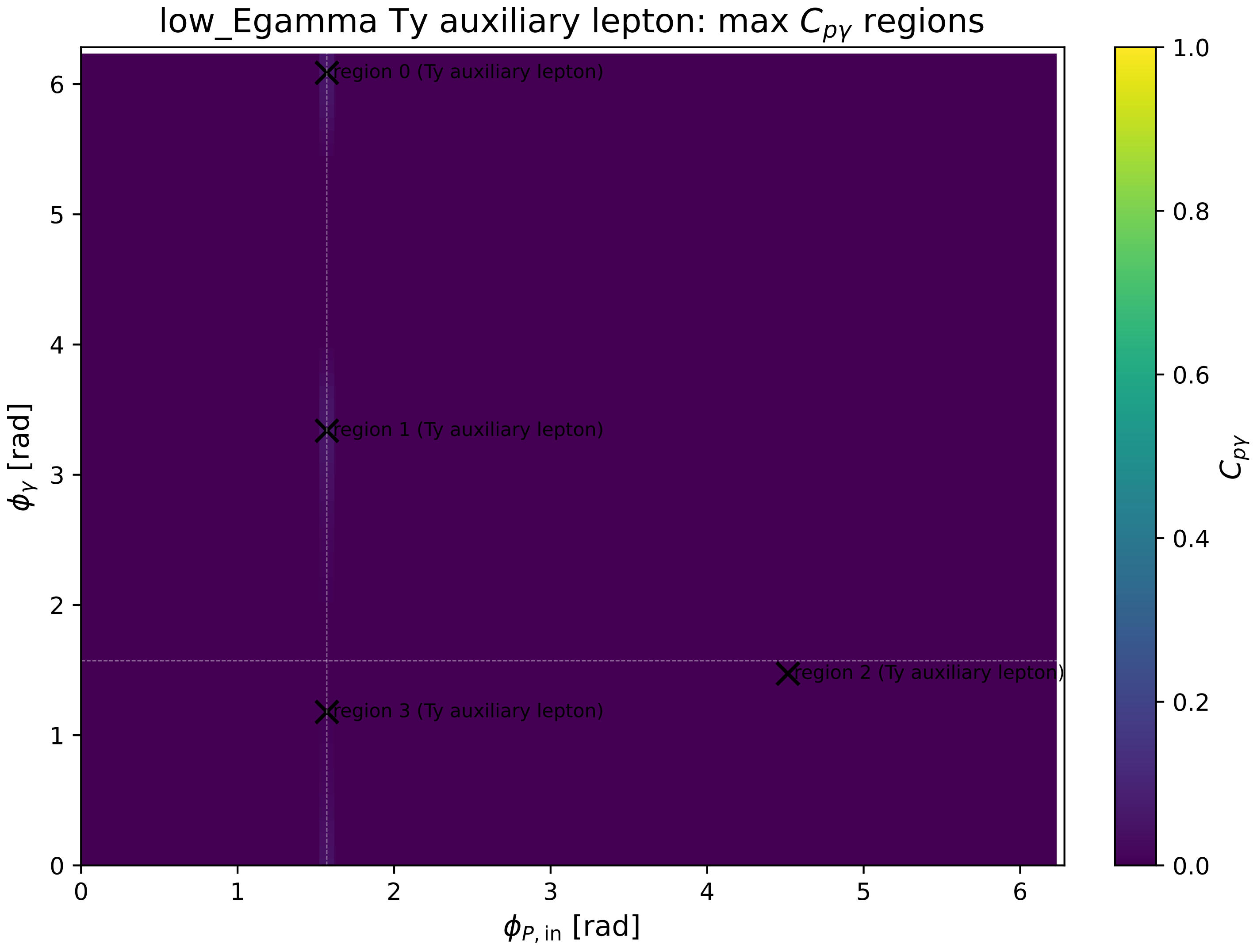
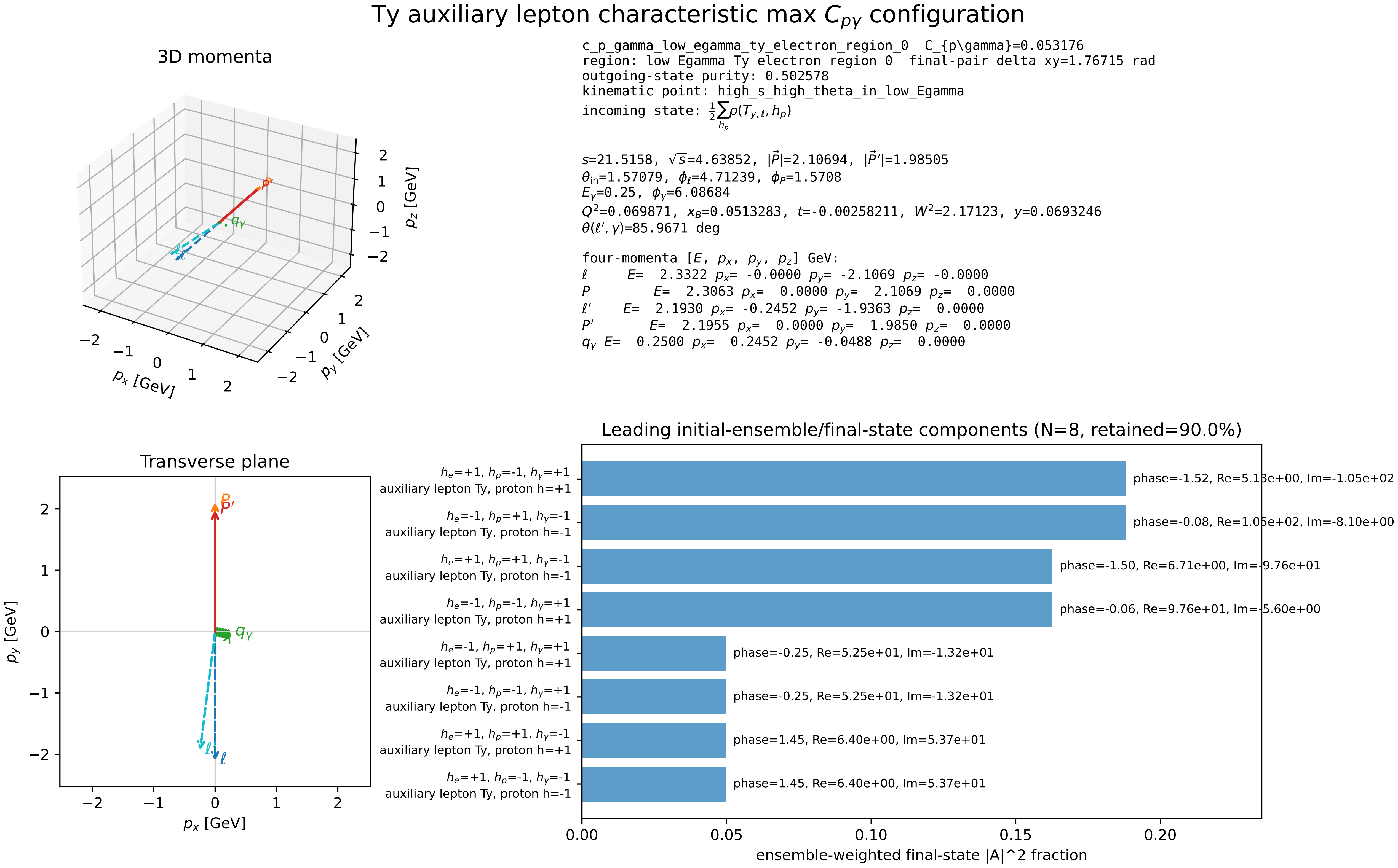
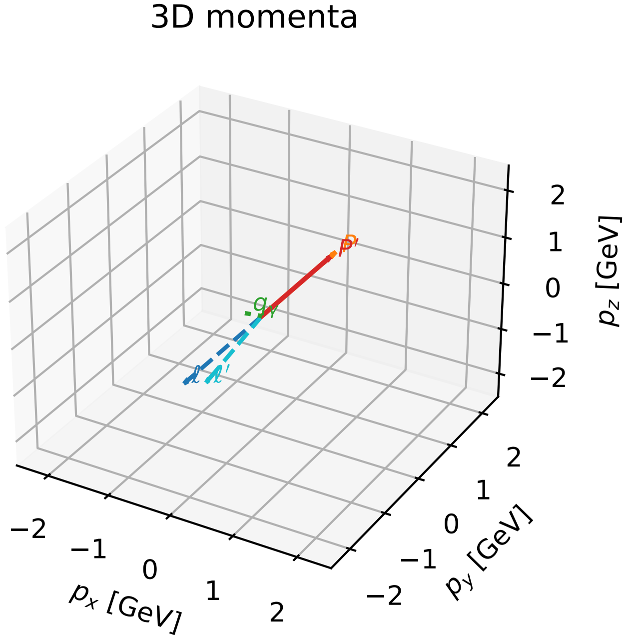


low_Egamma Ty auxiliary lepton: max $C_{p\gamma}$ regions
region 0 (Ty auxiliary lepton)
region 1 (Ty auxiliary lepton)
region 2 (Ty auxiliary lepton)
region 3 (Ty auxiliary lepton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{p\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0


Ty auxiliary lepton characteristic max $C_{p\gamma}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
q_γ
ℓ'
ℓ
c_p_gamma_low_egamma_ty_electron_region_0 C_{p\gamma}=0.053176
region: low_Egamma_Ty_electron_region_0 final-pair delta_xy=1.76715 rad
outgoing-state purity: 0.502578
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.98505
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=6.08684
Q^2=0.069871, x_B=0.0513283, t=-0.00258211, W^2=2.17123, y=0.0693246
$\theta(\ell', \gamma)$=85.9671 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.0000 p_y= -2.1069 p_z= -0.0000
P E= 2.3063 p_x= 0.0000 p_y= 2.1069 p_z= 0.0000
ℓ' E= 2.1930 p_x= -0.2452 p_y= -1.9363 p_z= 0.0000
P' E= 2.1955 p_x= 0.0000 p_y= 1.9850 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2452 p_y= -0.0488 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
P
P'
q_γ
ℓ'
ℓ
Leading initial-ensemble/final-state components (N=8, retained=90.0%)
h_e=+1, h_p=-1, h_γ=+1
auxiliary lepton Ty, proton h=+1
phase=-1.52, Re=5.13e+00, Im=-1.05e+02
h_e=-1, h_p=+1, h_γ=-1
auxiliary lepton Ty, proton h=-1
phase=-0.08, Re=1.05e+02, Im=-8.10e+00
h_e=+1, h_p=+1, h_γ=-1
auxiliary lepton Ty, proton h=-1
phase=-1.50, Re=6.71e+00, Im=-9.76e+01
h_e=-1, h_p=-1, h_γ=+1
auxiliary lepton Ty, proton h=+1
phase=-0.06, Re=9.76e+01, Im=-5.60e+00
h_e=-1, h_p=+1, h_γ=+1
auxiliary lepton Ty, proton h=+1
phase=-0.25, Re=5.25e+01, Im=-1.32e+01
h_e=-1, h_p=-1, h_γ=+1
auxiliary lepton Ty, proton h=-1
phase=-0.25, Re=5.25e+01, Im=-1.32e+01
h_e=+1, h_p=+1, h_γ=-1
auxiliary lepton Ty, proton h=+1
phase=1.45, Re=6.40e+00, Im=5.37e+01
h_e=+1, h_p=-1, h_γ=-1
auxiliary lepton Ty, proton h=-1
phase=1.45, Re=6.40e+00, Im=5.37e+01
ensemble-weighted final-state |A|^2 fraction

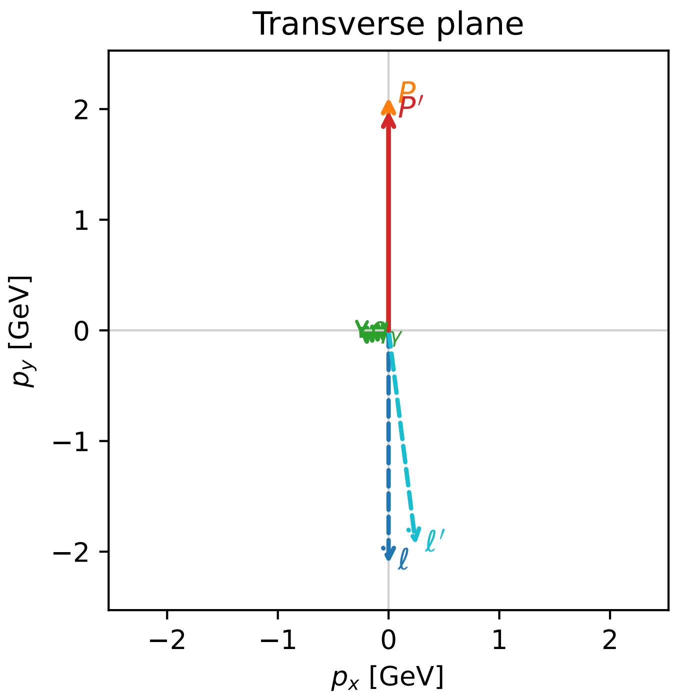
Ty auxiliary lepton characteristic max $C_{p\gamma}$ configuration

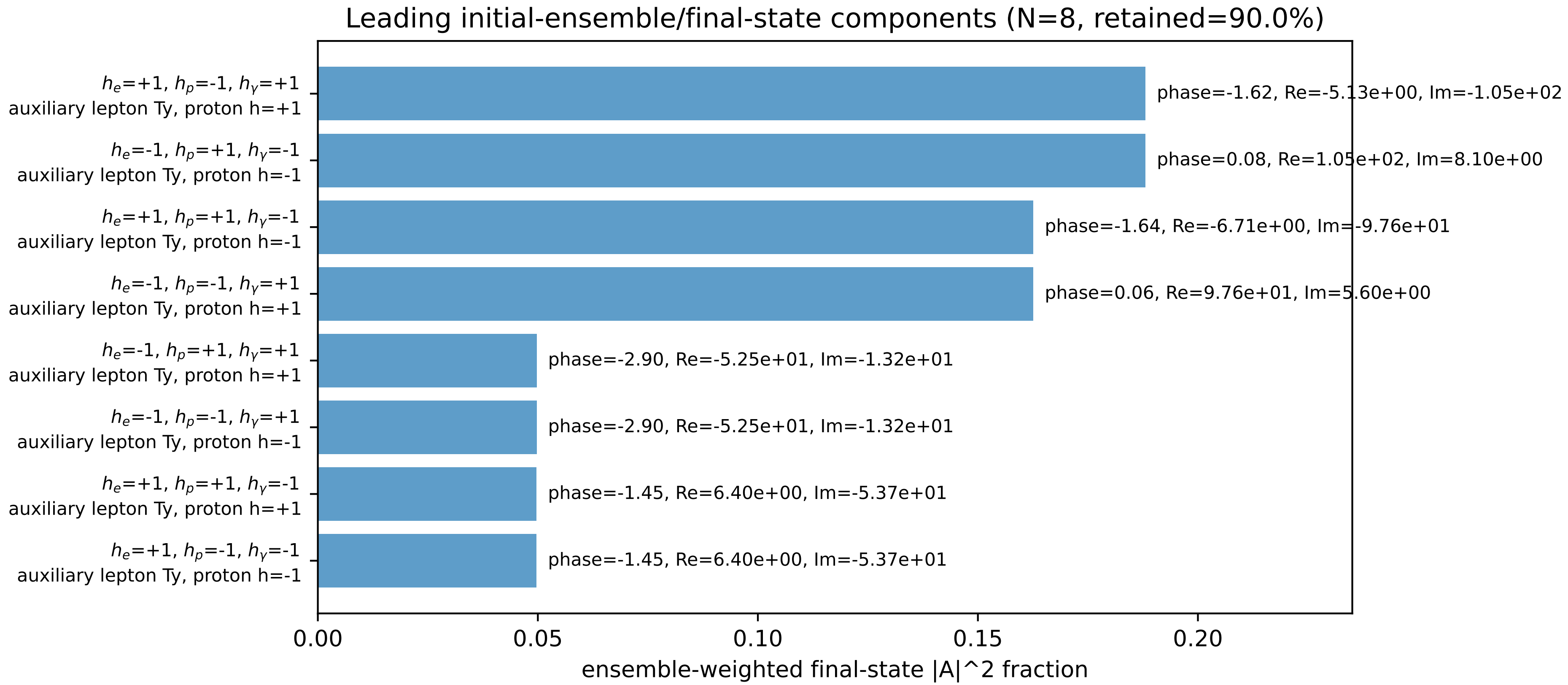


c_p_gamma_low_egamma_ty_electron_region_1 C_{p\gamma}=0.053176
region: low_Egamma_Ty_electron_region_1 final-pair delta_xy=1.76715 rad
outgoing-state purity: 0.502578
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell},h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.98505
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=3.33794
Q^2=0.069871, x_B=0.0513283, t=-0.00258211, W^2=2.17123, y=0.0693246
$\theta(\ell',\gamma)$=85.9671 deg

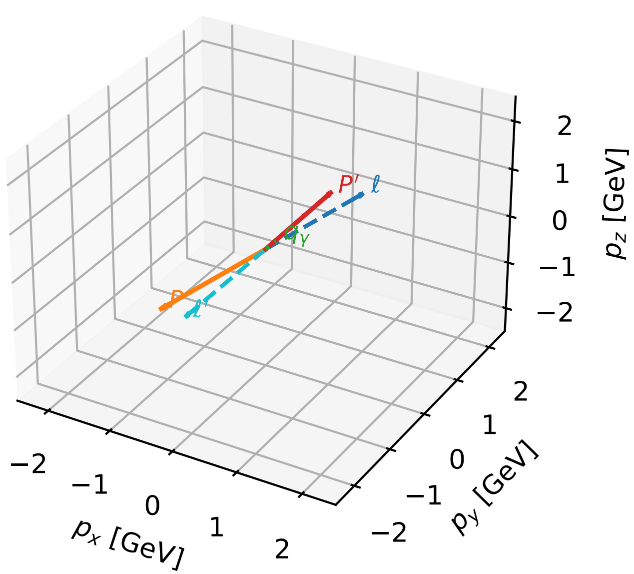
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.0000 p_y= -2.1069 p_z= -0.0000
P E= 2.3063 p_x= 0.0000 p_y= 2.1069 p_z= 0.0000
ℓ' E= 2.1930 p_x= 0.2452 p_y= -1.9363 p_z= 0.0000
P' E= 2.1955 p_x= 0.0000 p_y= 1.9850 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2452 p_y= -0.0488 p_z= 0.0000





Ty auxiliary lepton characteristic max $C_{p\gamma}$ configuration

3D momenta




```
c_p_gamma_low_egamma_ty_electron_region_2  C_{p\gamma}=0.00331056
region: low_Egamma_Ty_electron_region_2  final-pair delta_xy=0.0981748 rad
outgoing-state purity: 0.504148
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.84405

θ_{in}=1.57079, ϕ_ℓ=1.37445, ϕ_P=4.51604

E_γ=0.25, ϕ_γ=1.47262

Q^2=17.4894, x_B=0.993362, t=-15.4047, W^2=0.996711, y=0.89663

$\theta(\ell', \gamma)$=175.046 deg

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ E= 2.3322 p_x= 0.4110 p_y= 2.0665 p_z= -0.0000

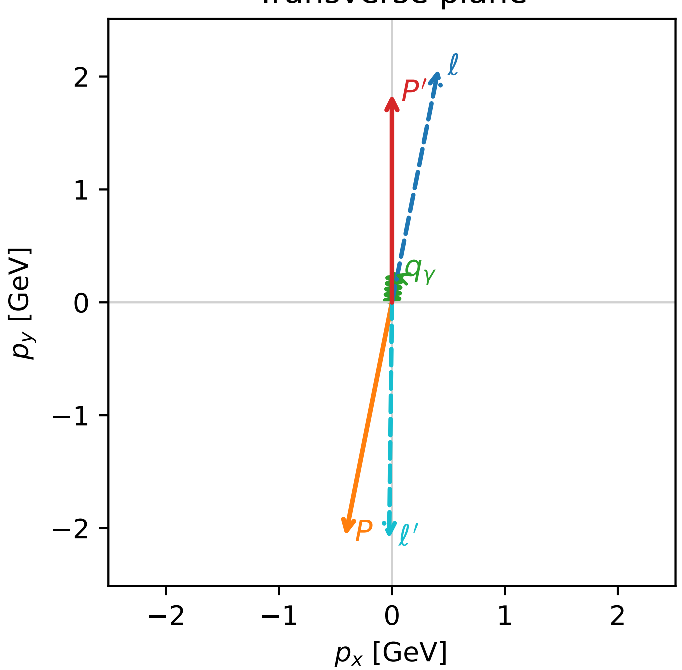
P E= 2.3063 p_x= -0.4110 p_y= -2.0665 p_z= 0.0000

ℓ' E= 2.3196 p_x= -0.0245 p_y= -2.0928 p_z= 0.0000

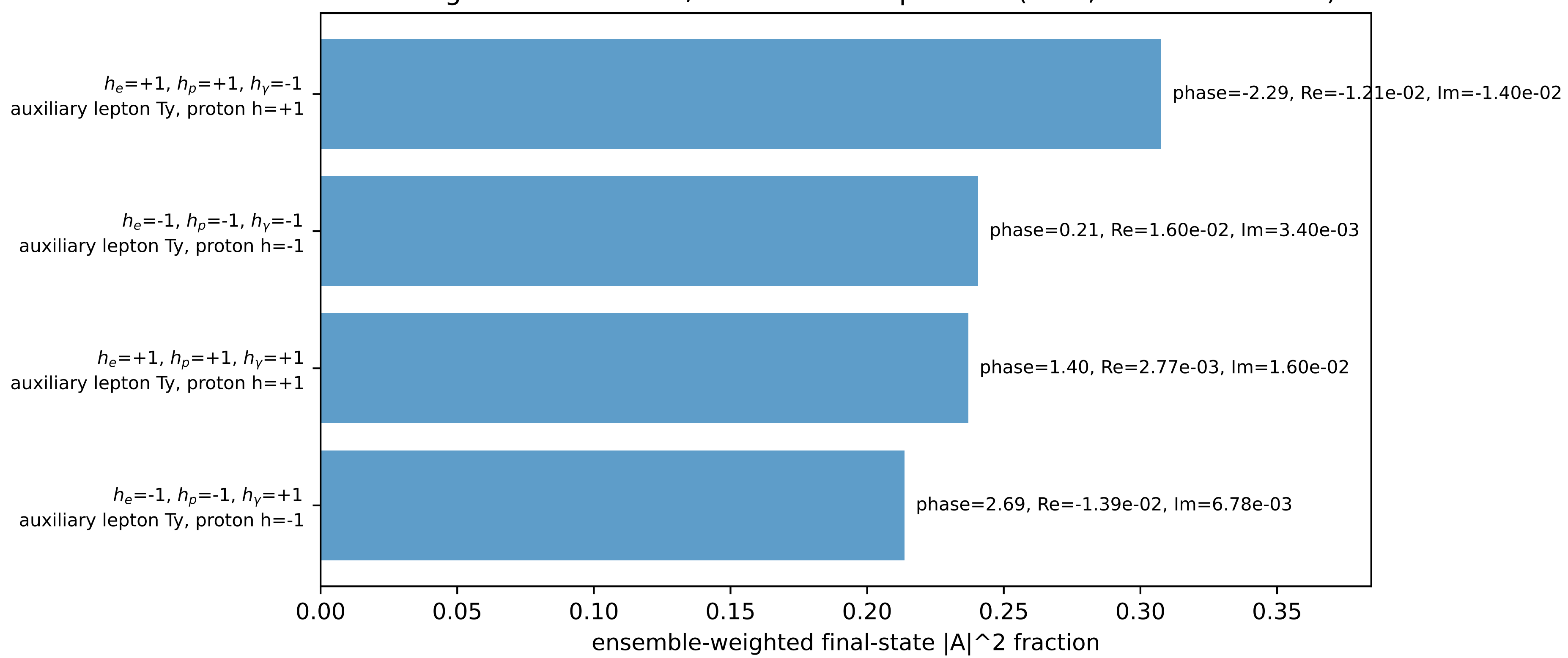
P' E= 2.0689 p_x= 0.0000 p_y= 1.8441 p_z= 0.0000

q_γ E= 0.2500 p_x= 0.0245 p_y= 0.2488 p_z= 0.0000

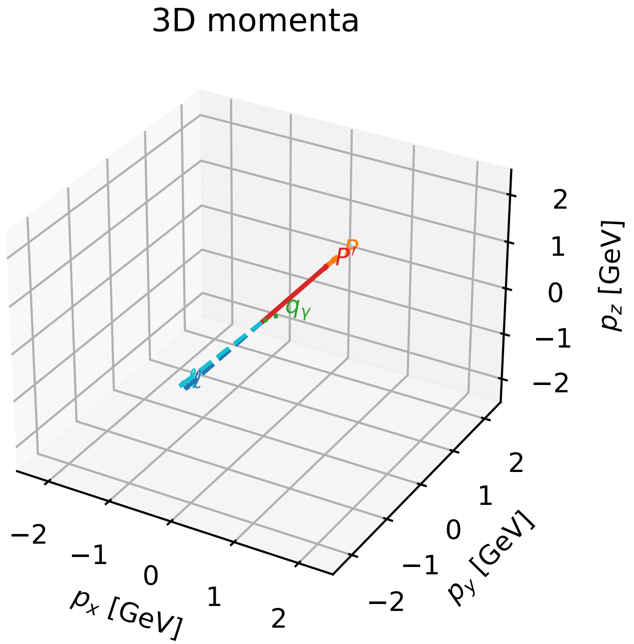
Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.9%)



Ty auxiliary lepton characteristic max $C_{p\gamma}$ configuration



c_p_gamma_low_egamma_ty_electron_region_3 C_{p\gamma}=0.00250801
region: low_Egamma_Ty_electron_region_3 final-pair delta_xy=0.392699 rad
outgoing-state purity: 0.500004
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.85198
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=1.1781
Q^2=0.00934153, x_B=0.0486979, t=-0.0119532, W^2=1.06233, y=0.0097691
$\theta(\ell', \gamma)$=160.13 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.0000 p_y= -2.1069 p_z= -0.0000
P E= 2.3063 p_x= 0.0000 p_y= 2.1069 p_z= 0.0000
ℓ' E= 2.3125 p_x= -0.0957 p_y= -2.0830 p_z= 0.0000
P' E= 2.0760 p_x= 0.0000 p_y= 1.8520 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0957 p_y= 0.2310 p_z= 0.0000